

LAB REPORT: ASSIGNMENT 01

Topic: Dataset Exploration using Python & R

- **Course Name:** Introduction To Data Science
- **Course Code :** ADS 201
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1. Objective

The objective of this assignment is to perform basic data exploration on the `Space_Corrected.csv` dataset using both Python (Pandas) and R programming languages, and compare the steps involved in loading, inspecting, and summarizing data.

2. Dataset Overview

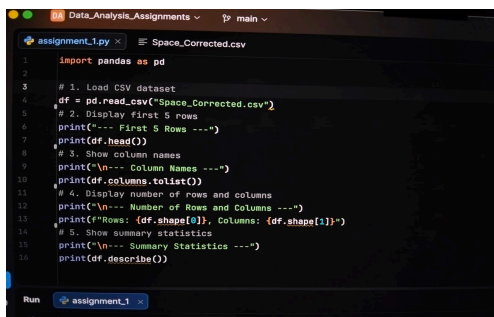
- **Dataset Name:** Space_Corrected.csv
- **Description:** Historical data on space mission launches containing organization names, location, date, rocket details, status, and cost.

3. Part A: Implementation in Python

Steps Taken:

1. Loaded the dataset using the `pandas` library.
2. Inspected the first 5 rows using `df.head()`.
3. Checked column names, data types, and shape using `df.info()` and `df.shape`.
4. Generated descriptive statistics using `df.describe()`.

Python Code:

A screenshot of a Jupyter Notebook interface. The top bar shows 'Data_Analysis_Assignments' and 'main'. The notebook has two tabs: 'assignment_1.py' (active) and 'Space_Corrected.csv'. The code in the cell is as follows:

```
1 import pandas as pd
2
3 # 1. Load CSV dataset
4 df = pd.read_csv("Space_Corrected.csv")
5 # 2. Display first 5 rows
6 print("--- First 5 Rows ---")
7 print(df.head())
8 # 3. Show column names
9 print("\n--- Column Names ---")
10 print(df.columns.tolist())
11 # 4. Display number of rows and columns
12 print("\n--- Number of Rows and Columns ---")
13 print(f"Rows: {df.shape[0]}, Columns: {df.shape[1]}")
14 # 5. Show summary statistics
15 print("\n--- Summary Statistics ---")
16 print(df.describe())
```

The bottom of the notebook shows a 'Run' button and a status bar with 'assignment_1' and 'C2'.

Output & Observations:

```
Run assignment_1 x
4 4 4 ULA ... StatusActive 145.0 Success
[5 rows x 9 columns]
--- Column Names ---
['Unnamed: 0.1', 'Unnamed: 0', 'Company Name', 'Location', 'Datum', 'Detail', 'Status Rocket', 'Rocket', 'Status Mission']
```

```
Run assignment_1 x
--- Number of Rows and Columns ---
Rows: 4324, Columns: 9

--- Summary Statistics ---
                Unnamed: 0.1  Unnamed: 0
count  4324.000000  4324.000000
mean    2161.500000  2161.500000
```

```
Run assignment_1 x
/Users/kitty/PycharmProjects/Data_Analysis_Assignments1/.venv/bin/python /Users/kitty/PycharmProjects/Data_Analysis_Assignments1/assignment_1.py
--- First 5 Rows ---
Unnamed: 0.1  Unnamed: 0  Company Name  ...  Status Rocket  Rocket Status Mission
0            0            0  SpaceX    ...  StatusActive  50.0      Success
1            1            1    CASC     ...  StatusActive  29.75     Success
2            2            2  SpaceX    ...  StatusActive  NaN       Success
3            3            3  Roscosmos ...  StatusActive  65.0      Success
4            4            4    ULA      ...  StatusActive  145.0     Success
```

```
Run assignment_1 x
min            0.000000  0.000000
25%          1080.750000  1080.750000
50%          2161.500000  2161.500000
75%          3242.250000  3242.250000
max          4323.000000  4323.000000

Process finished with exit code 0
```

```
Run assignment_1 x
mean          2161.500000  2161.500000
std           1248.375611  1248.375611
min            0.000000  0.000000
25%          1080.750000  1080.750000
50%          2161.500000  2161.500000
75%          3242.250000  3242.250000
max          4323.000000  4323.000000
```

4. Part B: Implementation in R

Steps Taken:

1. Set up an R project in RStudio and loaded the CSV dataset using `read.csv()`.
2. Inspected the first 5 records with `head(df, 5)`.
3. Extracted column headers using `colnames(df)` and verified dimensions (4,324 rows and 9 columns) using `dim(df)`.
4. Obtained a summary of all variables using `summary(df)`.

R Code:

```
assignment_1.R*  
1 # 1. Load CSV Dataset  
2 df <- read.csv("Space_Corrected.csv")  
3  
4 # 2. First 5 rows  
5 head(df, 5)  
6  
7 # 3. Column names  
8 colnames(df)  
9  
10 # 4. Total rows and columns  
11 dim(df)  
12  
13 # 5. Summary statistics  
14 summary(df)
```

Output & Observations:

```
R - R 4.6.1 - ~/PycharmProjects/Data_Analysis_Assignments1/  
> # 1. Load CSV Dataset  
> df <- read.csv("Space_Corrected.csv")  
>  
> # 2. First 5 rows  
> head(df, 5)  
  X Unnamed..0 Company.Name Location  
1 0 0 SpaceX LC-39A, Kennedy Space Center, Florida, USA  
2 1 1 CASC Site 9401 (SLS-2), Jiuquan Satellite Launch Center, China  
3 2 2 SpaceX Pad A, Boca Chica, Texas, USA  
4 3 3 Roscosmos Site 200/39, Baikonur Cosmodrome, Kazakhstan  
5 4 4 ULA SLC-41, Cape Canaveral AFS, Florida, USA  
  Datum Detail Status.Rocket Rocket  
1 Fri Aug 07, 2020 05:12 UTC Falcon 9 Block 5 | Starlink V1 L9 & BlackSky StatusActive 50.0  
2 Thu Aug 06, 2020 04:01 UTC Long March 2D | Gaofen-9 04 & Q-SAT StatusActive 29.75
```

```
Console Terminal Background Jobs  
R - R 4.6.1 - ~/PycharmProjects/Data_Analysis_Assignments1/  
2 Thu Aug 06, 2020 04:01 UTC Long March 2D | Gaofen-9 04 & Q-SAT StatusActive 29.75  
3 Tue Aug 04, 2020 23:57 UTC Starship Prototype | 150 Meter Hop StatusActive  
4 Thu Jul 30, 2020 21:25 UTC Proton-M/Briz-M | Ekspress-80 & Ekspress-103 StatusActive 65.0  
5 Thu Jul 30, 2020 11:50 UTC Atlas V 541 | Perseverance StatusActive 145.0  
Status.Mission  
1 Success  
2 Success  
3 Success  
4 Success  
5 Success  
>  
> # 3. Column names  
> colnames(df)  
[1] "X" "Unnamed..0" "Company.Name" "Location" "Datum"
```

```
Console Terminal Background Jobs R Script  
R - R 4.6.1 - ~/PycharmProjects/Data_Analysis_Assignments1/  
> # 3. Column names  
> colnames(df)  
[1] "X" "Unnamed..0" "Company.Name" "Location" "Datum"  
[6] "Detail" "Status.Rocket" "Rocket" "Status.Mission"  
>  
> # 4. Total rows and columns  
> dim(df)  
[1] 4324 9  
>  
> # 5. Summary statistics  
> summary(df)  
  X Unnamed..0 Company.Name Location Datum Detail  
Min. : 0 Min. : 0 Length :4324 Length :4324 Length :4324 Length :4324  
1st Qu.:1081 1st Qu.:1081 N.unique : 56 N.unique : 137 N.unique :4319 N.unique :4278
```

```
Console Terminal Background Jobs
R 4.6.1 ~/PycharmProjects/Data_Analysis_Assignments1/
Min. : 0 Min. : 0 Length :4324 Length :4324 Length :4324 Length :4324
1st Qu.:1081 1st Qu.:1081 N.unique : 56 N.unique : 137 N.unique :4319 N.unique :4278
Median :2162 Median :2162 N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0
Mean :2162 Mean :2162 Min.nchar: 2 Min.nchar: 25 Min.nchar: 16 Min.nchar: 10
3rd Qu.:3242 3rd Qu.:3242 Max.nchar: 16 Max.nchar: 87 Max.nchar: 26 Max.nchar: 70
Max. :4323 Max. :4323
Status.Rocket Rocket Status.Mission
Length :4324 Length :4324 Length :4324
N.unique : 2 N.unique : 57 N.unique : 4
N.blank : 0 N.blank :3360 N.blank : 0
Min.nchar: 12 Min.nchar: 0 Min.nchar: 7
Max.nchar: 13 Max.nchar: 8 Max.nchar: 17
```

5. Conclusion

Both Python and R successfully loaded and explored the space launch dataset. The dataset consists of 4,324 rows and 9 columns. Both environments provided equivalent summary metrics and data structural details, demonstrating effective data exploration capability in both languages.